FundusShift and LOTUS: Benchmarking and Improving Fundus Disease Classification at Unseen Sources

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Abstract. A fundus classifier that excels on validation may fail on images from a new source, and so may its confidence scores. FundusShift is a leakage-controlled benchmark of normal, diabetic retinopathy (DR) and glaucoma images from fifteen public collections, with seven held-out test sources and a separate glaucoma-screening set. Under one protocol we evaluate 818 configurations of fine-tuned, frozen, prompt-based and domain generalization models, plus four multimodal LLMs, and repeat the key analyses over 50 random source rotations. Three findings stand out. Validation scores barely rank fine-tuned models at unseen sources (Spearman 0.05). Thresholds and conformal quantiles tuned on development data break their promises at new sources. Glaucoma, present in only two development sources, is the weakest class. We then propose LOTUS, leave-one-source-out linear heads on frozen MedSigLIP tokens fused with zero-shot text scores (about 8,000 trained parameters). Over the 50 rotations, with three seeds each, LOTUS reached a balanced accuracy of 0.789 against 0.746 for the strongest probe (higher on 43 of 50) and an expected calibration error of 0.034 against 0.084 (lower on 47 of 50), and it was also the most accurate on average when whole regions were held out. Its ablation shows that text fusion carries the calibration gain and that removing between-site directions, part of our initial design, lowers accuracy, because those directions also carry disease signal. Fitting thresholds on leave-one-source-out predictions kept a 10% risk target on 42 of 50 rotations instead of 19. Fundus benchmarks should report results per source and class, and check reliability promises at unseen sources. **Code, results and supplement:** https://github.com/dangaura-tech/fundusshift (archived at https://doi.org/10.5281/zenodo.23979471)

Keywords: Fundus photography · Source shift · Benchmark · Calibration · Foundation models · Glaucoma.

1 Introduction

Diabetic retinopathy (DR) and glaucoma affect large populations worldwide [46,47], and fundus photography offers a low-cost way to screen for them. Deep models reach high accuracy within a study population [13], yet performance can drop at new sites [53]. A screening tool must also *know* when it may be wrong, but its thresholds and confidence scores are tuned during development, and uncertainty degrades under dataset shift [34].

Evidence for fundus images leaves three gaps. Domain generalization (DG) studies usually target one disease and measure accuracy alone [2,7]. General DG bench-

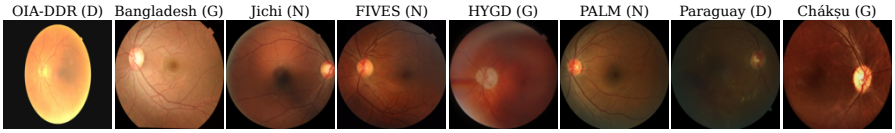


Fig. 1. Images from eight public sources (class in brackets), showing differences in field of view, colour and quality.

marks found that specialized algorithms rarely beat well-tuned empirical risk minimization [12,21]. Foundation models [59,42,41] are often evaluated where pretraining may overlap with test data; we found such overlap for FLAIR on five of our eight held-out sources. No shared protocol answers a practical question: *which approach, if any, keeps both its accuracy and its reliability at a source it has never seen?*

Our contributions are:

- **FundusShift**, a three-class benchmark from fifteen public collections, with leakage groups, source-held-out test data, a separate glaucoma-screening set and dedicated calibration splits (Sec. 3).
- **A reliability-first evaluation** of 818 configurations (25 encoders, 16 training settings) and four multimodal LLMs, covering raw and calibrated scores, selective prediction, conformal sets and 50 source rotations (Sec. 4). Validation rank, calibration and risk control do not reliably transfer, and glaucoma drives much of the failure.
- **LOTUS**, a low-capacity method on a frozen encoder that is more accurate than every baseline over the rotations, under region hold-out and on the published split, and better calibrated than every trained baseline over the rotations (Secs. 5 and 6). A full factorial ablation shows what transfers (regularized linear heads on frozen tokens, text fusion) and why explicit removal of site directions does not.
- **A shift-aware reference for promises**: thresholds fitted on leave-one-source-out predictions instead of in-distribution calibration data keep a risk target far more often, at a known cost in coverage (Sec. 6.5).

2 Related Work

Domain generalization. DRGen [2] and GDRNet [7] address DR grading at unseen domains; we cover DR and glaucoma and measure reliability as well as accuracy. DomainBed and WILDS showed that model selection and realistic splits are decisive [12,21]; we follow them and include Group DRO [40], CORAL [44] and SWAD [6]. DAEL [58] ensembles one expert head per source; LOTUS instead fits each head *without* one source, on a frozen encoder.

Foundation and vision-language models. RETFound [59], FLAIR [42], BiomedCLIP [56] and MedSigLIP [41] supply pretrained encoders, alongside general ones such as DINOv2 [33] and SigLIP [54,49]. We also test prompt learning [57], adapters [55],

Table 1. FundusShift sources and images by class: normal (N), diabetic retinopathy (D), glaucoma (G). Development (Dev) sources are split into train, validation and calibration. A further 861 out-of-scope images form an open-set probe (Suppl. S7).

Role	Source	Region	N	D	G	Total
Dev	Bangladesh Eye [20]	South Asia	846	1,500	1,156	3,502
Dev	OIA-DDR [27]	East Asia	6,265	6,255	–	12,520
Dev	DeepDRiD [30]	East Asia	914	1,086	–	2,000
Dev	GRAPE [17]	East Asia	–	–	586	586
Dev	IDRiD [36]	South Asia	167	347	–	514
Dev	Jichi [45]	East Asia	6,295	3,120	–	9,415
Dev	SUSTech-SYSU [28]	East Asia	631	588	–	1,219
Test	AFIO [15]	South Asia	67	18	44	129
Test	FIVES [19]	East Asia	193	200	193	586
Test	HRF [4]	Europe	15	15	15	45
Test	HYGD [1]	Middle East	–	–	548	548
Test	PALM [9]	East Asia	161	–	–	161
Test	PAPILA [23]	Europe	333	–	87	420
Test	Paraguay [5]	Latin America	708	723	–	1,431
OOD	Chákṣu [25]	South Asia	1,156	–	187	1,343

weight interpolation [52] and multimodal LLMs [8]. Linear probing can outperform fine-tuning out of distribution [24]; we extend this observation to reliability.

Reliability under shift. Temperature scaling (TS) [14], focal temperature scaling (FTS) [22] and calibration-aware losses [16,29] target better probabilities. Selective classification [10] and conformal prediction [38] control risk and coverage under exchangeability, which a new site breaks [48]; calibrating on held-out training domains improves calibration under shift [50,11], which our leave-one-source-out reference applies to thresholds. Test-time adaptation [51,18] and prior-shift estimation [39] correct for shift without labels.

Nuisance removal. Nuisance attribute projection removes channel directions in speaker recognition [43], like linear concept erasure [37,3]. Our initial design adapted it to acquisition sites; we show that it erases task-correlated directions here and drop it. Each component of LOTUS is known; we did not find their combination in fundus work (our search was not systematic).

3 The FundusShift Benchmark

Data. Table 1 lists the sources, all publicly released; this is a secondary analysis under each source's licence, with labels used as published. Development data comprise 23,805 training, 2,983 validation and 2,968 calibration images. The test split holds 3,320 images from seven unseen sources, and Chákṣu (1,343 images) is kept apart as a glaucoma-screening view. A source is one public collection: all but OIA-DDR (147 hospitals) come from a single institution or study, so held-out units are collections, not hospitals, and no acquisition institution spans development and test (Suppl. S1). Two structural facts shape the results. G appears in only two development sources, and GRAPE holds nothing else, so a model can learn their appearance instead of the disease. Six of seven development sources graded only DR, so their "normal" means "no DR" (Suppl. S1).

Table 2. Coverage of the benchmark on test, FLAIR excluded (reported apart). Configs: distinct (setting, encoder, configuration). Runs: including seeds. Selection: what validation chooses.

Setting	Encoders and options	Configs	Runs	Selection
Fine-tuning	13 CNN/ViT/RETFound [59] × 5 losses [16,29]	65	195	best epoch
Fine-tuning, NLL rule	Sensitivity check	2	6	best epoch, NLL
Linear probe	25 encoders, 3 weightings, 5 losses	375	1125	best epoch
MLP probe	Frozen encoders	11	33	best epoch
16-shot probe	Frozen encoders, 16 per class	125	375	best epoch
Zero-shot	Vision-language encoders, 3 prompt banks	33	33	prompt bank
CoOp [57] and variants	Vision-language encoders	190	570	fixed recipe
Tip-Adapter [55]	Vision-language encoders	6	18	α, β, bank
ERM, Group DRO [40], CORAL [44], SWAD [6]	ConvNeXt-T [32], DINOv2 LoRA [33]	7	21	best epoch; SWAD: own rule
WiSE-FT [52]	Vision-language encoders	4	12	α, checkpoint
Multiple choice	4 multimodal LLMs, e.g. [8], 6 option orders	4	4	none

Two test sources hold a single class (HYGD, PALM), for which bAcc reduces to recall; Sec. 6.3 reports results without them.

Leakage control. Exact hashes, vessel-pattern normalized cross-correlation and patient identifiers are joined into leakage groups, which never span two splits. This found, among others, brightness-altered copies of GRAPE photographs inside FIVES, which were removed. An automated audit confirms that no leakage group, file, patient, near-duplicate pair or acquisition family spans the development and test sides (Suppl. S1).

Protocol. Validation selects checkpoints and hyperparameters; calibration subset Cal-A fits temperatures; Cal-B fits thresholds and conformal quantiles; test and Chákṣu are used only for reporting. Each setting's *representative* is its configuration with the highest validation balanced accuracy (bAcc), so test scores play no part in choosing baselines. Raw, TS and FTS scores pass through identical code. FLAIR is reported apart, since its pretraining names FIVES, HRF, PAPILA, Paraguay and Chákṣu.

Methods. Table 2 summarizes the benchmark. Trained settings use three seeds; fine-tuning losses are cross-entropy, label smoothing, focal, MDCA [16] and MbLS [29]. Recipes were audited against pitfalls such as input-size mismatches (Suppl. S2).

Metrics and statistics. We report pooled bAcc, equal-source bAcc (Eq-src), macro AUROC and class recalls; ECE with 15 bins [35], NLL and Brier score; and coverage, risk and false reassurance (disease answered normal) at thresholds frozen before testing. Paired comparisons use 2,000 bootstrap draws over leakage groups with Holm correction. We also retrain on 50 random 7/7 partitions of the 14 non-OOD sources (80/10/10 group split of the development side); these share sources, so we report means and win counts, not p-values.

4 Diagnosis: What Breaks at an Unseen Source

Finding 1: Validation is a weak guide to new sources. Validation exceeded test bAcc in 94.5% of configurations, with a median gap of 0.194 (Fig. 2, left). The Spearman correlation between validation and test bAcc was 0.786 overall, but this mostly reflects differences between families: within fine-tuning, whose validation scores span 0.85–0.89, it was 0.053, and on Chákṣu it was negative (−0.156). The MedSigLIP linear probe had the highest validation bAcc among representatives (0.920) but reached 0.728 on test and 0.347 on Chákṣu; fine-tuned Swin-T fell from 0.893 to 0.655 and 0.298.

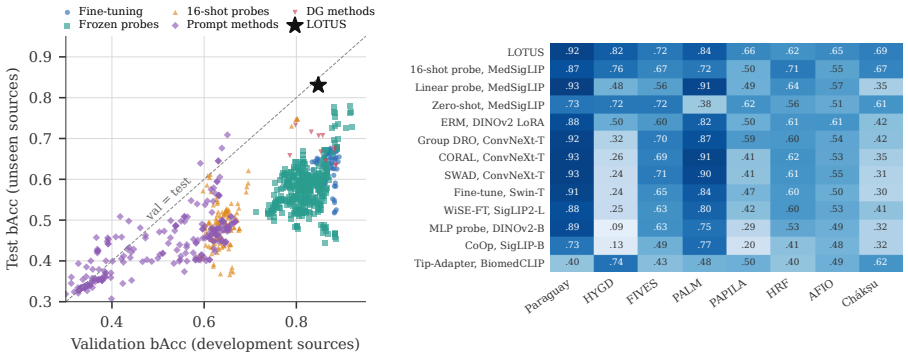


Fig. 2. Left: validation against test bAcc for 816 configurations (FLAIR, multimodal LLMs and the two NLL-rule sensitivity runs excluded). Most points lie well below the diagonal, and the fine-tuning cluster is nearly flat. Right: bAcc per unseen source for LOTUS (Sec. 5) and the validation-selected representatives; single-class sources show recall.

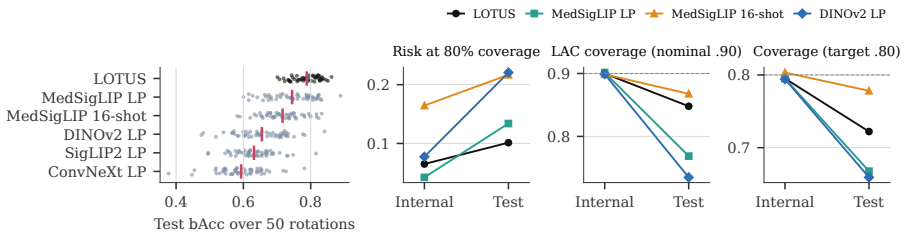


Fig. 3. Fifty source rotations. Left: test bAcc per rotation, mean as a red bar; LP: linear probe. Right: internal estimates on held-back Cal-B against unseen test sources, means over rotations, after TS. Dashed lines mark the nominal target.

Finding 2: Reliability does not transfer. Over 50 rotations and eight probe configurations (400 runs; Fig. 3, right; Suppl. S6), validation exceeded test in 98% of runs, risk at the threshold for 80% coverage exceeded its internal estimate in 89%, and coverage of 90% least-ambiguous (LAC) conformal sets [38] fell below its internal estimate in 85%. For the MedSigLIP linear probe, risk rose from 0.042 to 0.134 and LAC coverage fell from 0.901 to 0.769. Splitting Cal-B in half shows that the promises hold on unseen halves of the same sources, so the failure comes from the new sources, not from a small calibration set (Suppl. S5). TS can harm an already calibrated model (linear probe: 0.024 to 0.079), FTS rarely improved on TS (Suppl. S3), and a class-prior predictor has ECE 0.063, so ECE must be read with accuracy.

Finding 3: Glaucoma is the weakest link. Averaged over test sources, pairwise AUROC was 0.90–0.95 for N–D and D–G across the main representatives, but only 0.64–0.75 for N–G (Suppl. S6). The difficulty is tied to shift rather than to the images alone: a logistic regression on frozen features reached a much higher bAcc when trained within each target source than when trained on development data (Table 3), although

Table 3. Trained on development sources against trained within the target source (cross-validated), logistic regression on frozen features, mean bAcc over nine encoders. Within-source learning can use source-specific cues, so it is a reference level, not a strict upper bound.

	AFIO	Chákṣu	FIVES	HRF	PAPILA	Paraguay
Trained on development	.516	.352	.581	.583	.394	.817
Trained within source	.635	.718	.857	.779	.757	.928

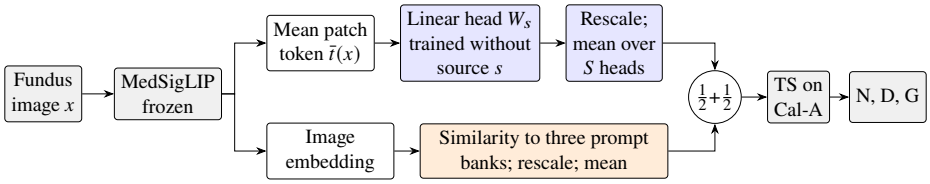


Fig. 4. LOTUS (Leave-One-source-out Token heads Unified with zero-shot Scores). Only the linear heads are trained, about 8,000 parameters in all; the encoder (about 0.4B parameters) and prompt banks are fixed. Low capacity refers to the trained part only.

AFIO stays low even within source. Site diversity is the bottleneck: over the rotations, recall G of the MedSigLIP linear probe rose from 0.468 with two glaucoma sources on the development side to 0.674 with four (LOTUS: 0.614 to 0.723). Models thus miss glaucoma or over-call it (Suppl. S4).

Finding 4: There is no cheap fix. DG losses did not close the gap: on ConvNeXt-T, Group DRO, CORAL and SWAD reached 0.707, 0.676 and 0.672 on test. Prompt-learning and adapter models (0.48–0.50) fell below plain zero-shot (0.709). Test-time adaptation (TENT [51], T3A [18]) gave no stable gain, and label-free prior estimation [39] was unstable (linear probe 0.835, zero-shot 0.635). Oracle per-source priors raised most representatives to 0.81–0.86, so class balance is a large part of the shift (Suppl. S7). Multimodal LLMs scored lowest (best: Fundus-R1 [8], 0.576).

A remedy must therefore generalize without target data and avoid fitting the appearance of the two glaucoma sources.

5 LOTUS

LOTUS (Fig. 4) follows one principle: *where site support is thin, spend little capacity, and do not depend on any single source.* A linear head on frozen features cannot memorize much site appearance, bagging over left-out sources avoids depending on any one of them, and text scores do not depend on the training sites' class balance.

Token input. MedSigLIP [41] takes a 448×448 image. We pool its last-layer patch tokens to a 14×14 grid, apply one fixed random orthonormal map to 384 dimensions (fitted on no data; it saved storage during development), and average over the grid to obtain $\bar{t}(x)$, standardized on training rows.

Leave-one-source-out bag. For each development source s whose removal leaves all three classes ($S = 7$ on the published split), we fit a linear head W_s (1,155 parameters) on all other sources, with label smoothing 0.1, class-balanced sampling, AdamW (learning

Table 4. Development ablation of the initial design (leave-one-development-source-out, three seeds, equal-site mean; no test data). False G: non-glaucoma images predicted G (not recorded for text only).

Configuration	bAcc	ECE	NLL	Recall G	False G
Single linear head on tokens	.689	.128	.717	.608	.164
Text only (three banks)	.758	.245	.749	.605	–
Bag + projection	.744	.110	.635	.825	.181
Single head + text	.759	.133	.616	.650	.098
Initial design: bag + projection + text	.792	.151	.575	.764	.100

rate 3×10^{-4}, weight decay 10^{-2}), batch size 64 and 12 epochs, keeping the epoch with the best validation bAcc on the head's own sites.

Text fusion. Three fixed prompt banks (a CLIP-style template ensemble, CuPL-style descriptions and one descriptive prompt; Suppl. S2) give zero-shot logits z_b^{text} from MedSigLIP's image embedding. Let $\mathcal{R}$ centre a logit vector per image and divide it by its standard deviation over validation images. The final logits are

$$z(x) = \frac{1}{2} \cdot \frac{1}{S} \sum_{s=1}^{S} \mathcal{R}\big(W_s\, \bar{t}(x)\big) + \frac{1}{2} \cdot \frac{1}{3} \sum_{b=1}^{3} \mathcal{R}\big(z_b^{\text{text}}(x)\big). \tag{1}$$

$\mathcal{R}$ puts both parts on one scale, so the equal weight needs no fitting, and prediction uses no target-source data. A temperature T is fitted on Cal-A; averaging compresses logits, so $T < 1$ is expected ($T = 0.444$).

Initial design: between-site projection. Our initial design also removed, before each head, the directions along which development sites differ with class held fixed: with $\mu_{r,c}$ the mean of site r and class c on the head's training rows and μ_c the class mean, U_s holds the leading right singular vectors of the stacked $\mu_{r,c} - \mu_c$ (11–14 in practice), and each head sees $(I - U_s U_s^\top)\, \bar{t}(x)$.

Design and selection. About 120 configurations were compared by leave-one-source-out on the seven development sources (Suppl. S8). Heavier heads lost the G class at held-out sites (an RPAttn head [26] on three token depths had recall G of 0.008), which led to the low-capacity design; on development data the projection raised recall G (Table 4). Ten configurations reached a development bAcc of at least 0.791; all ten were evaluated on the published test split and the one with the highest test bAcc became the initial design (sixth on development; all ten reached 0.791–0.817, above the strongest validation-selected representative, 0.748; Suppl. S9). The rotation ablation of Sec. 6.2 then showed that the projection lowers accuracy, so LOTUS drops it and is otherwise unchanged. Both choices used held-out results, so the published-split, rotation and region numbers of LOTUS are not untouched estimates; we report the initial design alongside it.

6 Evaluating LOTUS

6.1 Source Rotations

LOTUS was retrained on each of the 50 partitions (Table 5, Fig. 3 left). With three seeds, mean bAcc was 0.789 (5th–95th percentile 0.725–0.841); seeds changed it by

Table 5. Fifty source rotations, same partitions for all methods, TS on each rotation's Cal-A, one training seed per rotation. Risk: at the threshold for 80% coverage fitted on Cal-B. LAC: coverage at nominal 0.90. LP: linear probe with class-weighted cross-entropy. Last four columns: LOTUS minus the row, and rotations where LOTUS is better. With three seeds, LOTUS against the MedSigLIP LP gives +0.043 bAcc (higher on 43/50) and lower ECE on 47/50 (Suppl. S6).

			Recall						LOTUS minus row			
Method	bAcc	Eq-src	N	D	G	ECE	Risk	LAC	ΔbAcc	Higher	ΔECE	Lower
LOTUS	.788	.766	.813	.845	.708	.034	.101	.848	–	–	–	–
MedSigLIP LP	.745	.725	.794	.803	.637	.084	.134	.769	+0.044	41/50	-0.050	47/50
MedSigLIP 16-shot	.716	.713	.685	.742	.723	.056	.217	.868	+0.072	49/50	-0.022	36/50
DINOv2 LP	.655	.650	.684	.750	.531	.118	.221	.735	+0.133	49/50	-0.084	49/50
SigLIP2 LP	.631	.617	.636	.709	.549	.141	.280	.728	+0.157	50/50	-0.108	50/50
ConvNeXt LP	.593	.609	.644	.710	.426	.130	.287	.730	+0.195	50/50	-0.097	50/50

Table 6. Ablation over the 50 rotations, same splits and recipe, TS, means over three seeds. Last two columns: LOTUS minus the row in bAcc, and rotations where LOTUS is higher. The full factorial (nine arms, with AUROC, AURC and risk) is in Suppl. S6.

Configuration	bAcc	Macro-F1	Rec. N	Rec. G	False G	ECE	NLL	Chákṣu	ΔbAcc	Higher
LOTUS (bag + text)	.789	.739	.813	.709	.070	.034	.507	.694	–	–
+ projection (initial design)	.771	.718	.796	.666	.081	.033	.526	.668	+0.018	45/50
without bag (one head + text)	.786	.737	.814	.695	.069	.035	.509	.688	+0.003	35/50
without text (bag)	.779	.720	.795	.725	.088	.050	.566	.598	+0.010	27/50
single head, no bag, no text	.770	.713	.792	.699	.086	.053	.577	.590	+0.019	35/50
text only	.713	.653	.693	.614	.131	.022	.611	.594	+0.076	50/50

0.004, partitions by 0.039. Against the strongest baseline, the MedSigLIP linear probe, LOTUS gained +0.043 ± 0.049 bAcc (higher on 43 of 50) with lower ECE after TS (0.034 against 0.084; lower on 47 of 50), lower risk at the 80%-coverage threshold (0.101 against 0.134) and higher LAC coverage (0.848 against 0.769). Gains over the 16-shot probe (+0.072, 49 of 50) and other encoders (+0.133 to +0.195, at least 49 of 50) were larger. The lead holds at every glaucoma diversity observed in more than one rotation: with two glaucoma sources on the development side, 0.777 against 0.676 for the probe, and with five, 0.788 against 0.772 (Suppl. S6). Holding out whole regions, LOTUS averaged 0.822 against 0.808 for the probe and 0.791 for the initial design, ahead on South Asia and the Middle East and within 0.013 elsewhere, though not better calibrated there (ECE 0.061 against 0.058; Suppl. S6).

6.2 What Carries the Gain

Table 6 isolates each component. (i) *Text fusion carries the calibration gain*: without it, ECE rose from 0.034 to 0.050 and NLL from 0.507 to 0.566, while bAcc fell by 0.010. (ii) *A regularized token head carries the accuracy gain*: a single head on tokens, without bag or text, already reached 0.770 against 0.745 for the linear probe on pooled features; both are linear, so the gain comes from the token input and stronger regularization, not

Table 7. Published split, means over three seeds; raw scores unless marked TS. Representatives are chosen by validation; LOTUS and the initial design are not untouched estimates (Sec. 5); their Val is the validation split of the development sources. Image-level intervals in the text are conditional on these seven sources. †FLAIR's pretraining includes five of these sources; reference only.

		Test accuracy			Test recall			Test calibration			Chákṣu	
Setting, encoder	Val	bAcc	Eq-src	AUROC	N	D	G	ECE	ECE TS	NLL TS	bAcc	Rec. G
LOTUS (ours)	.848	.830	.748	.939	.832	.905	.755	.201	.029	.457	.695	.561
+ projection (initial design)	.824	.815	.723	.932	.789	.904	.753	.175	.019	.482	.733	.822
16-shot probe, MedSigLIP [41]	.802	.748	.683	.915	.555	.906	.783	.096	.044	.644	.675	.752
Linear probe, MedSigLIP [41]	.920	.728	.654	.896	.800	.896	.488	.024	.079	.694	.347	.011
Zero-shot, MedSigLIP [41]	.650	.709	.606	.862	.594	.850	.682	.207	.032	.690	.615	.904
ERM, DINOv2 LoRA [33]	.857	.707	.646	.868	.620	.899	.603	.154	.115	.944	.419	.148
Group DRO [40], ConvNeXt-T [32]	.849	.707	.649	.889	.806	.871	.443	.082	.091	.926	.417	.045
Fine-tune (NLL rule), ConvNeXt-T [32]	.870	.677	.619	.864	.720	.883	.426	.185	.133	1.036	.324	.041
CORAL [44], ConvNeXt-T [32]	.883	.676	.623	.857	.729	.880	.419	.198	.136	1.005	.349	.034
SWAD [6], ConvNeXt-T [32]	.883	.672	.622	.866	.726	.887	.404	.208	.139	1.056	.309	.014
Fine-tune, Swin-T [31]	.893	.655	.600	.839	.655	.898	.411	.223	.171	1.174	.298	.037
WiSE-FT [52], SigLIP2-L [49]	.886	.634	.587	.825	.615	.871	.417	.236	.168	1.243	.406	.094
MLP probe, DINOv2-B [33]	.870	.598	.524	.807	.694	.841	.260	.263	.172	1.259	.318	.055
CoOp ctx-init [57], BiomedCLIP [56]	.642	.497	.447	.685	.540	.842	.109	.287	.150	1.189	.384	.064
CoOp + LP [57], SigLIP-B [54]	.668	.496	.463	.713	.493	.814	.180	.243	.176	1.180	.314	.262
Tip-Adapter-F [55], BiomedCLIP [56]	.666	.495	.453	.682	.303	.770	.412	.225	.147	1.099	.532	.471
CoOp [57], SigLIP-B [54]	.660	.491	.458	.706	.482	.803	.189	.236	.175	1.180	.318	.255
Tip-Adapter [55], BiomedCLIP [56]	.597	.478	.491	.674	.293	.396	.745	.332	.091	1.020	.617	.758
16-shot probe, FLAIR [42]†	.809	.750	.684	.857	.662	.882	.707	.134	.128	.915	.683	.686

from lower capacity. The bag adds little (+0.003, higher on 35 of 50) but never hurts. (iii) *Removing between-site directions does not transfer*: adding the projection of the initial design lowered bAcc by 0.018 (LOTUS higher on 45 of 50), recall G from 0.709 to 0.666 and macro-F1 by 0.021, at equal ECE.

The cause is measurable. On development data, the removed subspace held 55–87% of the norm of each class-mean difference (N–D, N–G, D–G), against 15–23% for a random subspace of equal size (Suppl. S6). Sites differ in disease severity and case mix within each label, so "between-site" directions are largely disease directions, and removing them discards signal. The projection also fails at its own goal: a site probe falls from 0.962 to 0.354 on development sources after projection, yet still identifies unseen test sources at 0.998, because new sites differ along new directions. Development leave-one-source-out had favoured the projection and misjudged the single head (0.689 there, Table 4), so Finding 1 applies to our own design process as well.

6.3 Published Split

On the published split (Table 7), LOTUS reached a pooled bAcc of 0.830 and an equal-source bAcc of 0.748, above every representative (nearest: the MedSigLIP 16-shot probe, difference 0.082, 95% CI 0.067–0.097, Holm-adjusted $p = 0.009$; Suppl. S9). The advantage comes from balance: recalls were 0.832, 0.905 and 0.755 for N, D and

Table 8. Reliability after TS on the published split; thresholds and quantiles fitted on Cal-B. Cov. 80: threshold for 80% coverage, with coverage, risk, and rates of glaucoma and DR images accepted as normal (over all images of the class). Risk 10: selective-risk bound of 10%. LAC: nominal 0.90, with mean set size. Chákṣu has no D images.

	Test								Chákṣu			
	Cov. 80				Risk 10		LAC		Cov. 80			LAC
Method	Cov.	Risk	G→N	D→N	Cov.	Risk	Cov.	Size	Cov.	Risk	G→N	Cov.
LOTUS	.691	.078	.054	.035	.732	.086	.883	1.14	.437	.055	.071	.902
16-shot, MedSigLIP	.753	.195	.012	.016	.353	.057	.912	1.61	.509	.307	.096	.929
Linear probe, MedSigLIP	.610	.129	.061	.022	.800	.178	.751	1.01	.308	.275	.203	.626
Zero-shot, MedSigLIP	.711	.224	.107	.052	.182	.033	.924	1.76	.689	.576	.043	.842
Group DRO, ConvNeXt-T	.738	.182	.089	.042	.664	.165	.769	1.13	.570	.249	.476	.788
ERM, DINOv2 LoRA	.715	.223	.067	.012	.689	.221	.722	1.08	.552	.314	.346	.679
Fine-tune, Swin-T	.694	.250	.147	.024	.752	.263	.676	1.06	.444	.484	.282	.559
WiSE-FT, SigLIP2-L	.641	.260	.088	.025	.661	.268	.668	1.10	.417	.280	.367	.707

G, whereas the 16-shot probe trades N (0.555) for G (0.783) and the fine-tuned and DG models miss G (0.40–0.60). LOTUS led the 16-shot probe on seven of eight views (Fig. 2, right); on HRF its G recall is zero on 15 glaucoma images. The ranking does not depend on the single-class or DR-only sources: without HYGD and PALM, LOTUS reached 0.794 against 0.698–0.772 for the six strongest representatives, and on the four sources that graded all three classes (FIVES, PAPILA, HRF, AFIO), 0.728 against 0.591–0.689 (Suppl. S4). On Chákṣu, which no design or selection step used, LOTUS had the highest bAcc of the representatives (0.695) with high specificity for glaucoma (0.866) but modest sensitivity (0.561); the initial design reached 0.733 there, trading specificity (0.688) for sensitivity (0.822; Suppl. S4).

6.4 Calibration and Selective Prediction

Raw LOTUS is underconfident because averaging compresses logits; TS corrects this (ECE 0.201 to 0.029), whereas fine-tuned Swin-T stays overconfident after TS (0.171; Suppl. S5). After TS, LOTUS had the lowest NLL of all representatives (0.457) and a lower ECE than each of them, significantly for all but the 16-shot probe (adjusted $p = 0.09$) and zero-shot ($p = 0.80$); the initial design is lower still (0.019; Suppl. S9). Per-source ECE averages 0.119 and exceeds 0.20 on HRF and AFIO, whose 45 and 129 images give intervals too wide to certify calibration either way (Suppl. S4).

In Table 8, the 10% risk target held on test (0.086) at a coverage of 0.732, against 0.182 and 0.353 for the only two representatives that also kept it (zero-shot and the 16-shot probe); the other five missed it (0.165–0.268). LAC covered 0.883 with 1.14 classes per set, against 1.61 for the 16-shot probe. On Chákṣu the risk target also held (0.067), which the initial design missed (0.162). Promises still weaken at new sources: the 80%-coverage threshold accepted 69% of test images and 44% on Chákṣu, and 5.4% of test glaucoma images were accepted as normal (16-shot probe: 1.2%). Per source,

Table 9. Promises on unseen sources over 50 rotations, LOTUS after TS. Left: threshold for 80% coverage. Middle: selective guaranteed risk (SGR [10], $\delta = 0.05$) threshold for risk $\leq 10\%$, with the rotations where test risk held and the rate of disease images answered normal (FR). Right: LAC coverage at nominal 0.90.

	Coverage target .80		Risk target .10				LAC .90	
Thresholds fitted on	Cov.	Risk	Cov.	Risk	Held	FR	Test	Chákṣu
Cal-B (in distribution)	.724	.102	.695	.105	19/50	.070	.849	.724
Cal-B (in distribution) + normal certificate	.596	.096	.575	.099	22/50	.037	–	–
LOSO reference	.833	.131	.592	.075	42/50	.046	.916	.897
LOSO reference + normal certificate	.665	.121	.556	.073	45/50	.036	–	–

the risk target held on three of seven test sources and on Chákṣu (Suppl. S5): LOTUS reduces the reliability problem but does not solve it.

6.5 Keeping Promises with a Shift-Aware Reference

Since in-distribution thresholds over-promise (Finding 2), we fit temperature, thresholds and conformal quantiles on a *leave-one-source-out reference* instead of Cal-B: every development image is scored by a LOTUS fitted without its source, so the reference has already met a source shift. An optional *normal certificate* answers "normal" only if $p(\mathrm{N})$ exceeds a threshold that keeps disease answered normal at most 2% in the reference. Selection rules were written before the run.

The reference keeps its promises far more often (Table 9). The 80% coverage target gave 0.833 against 0.724; LAC coverage was 0.916 against 0.849 on test and 0.897 against 0.724 on Chákṣu; and the 10% risk target held on 42 of 50 rotations against 19 (initial design: 44 against 23). The gain is a better placed threshold, not a better ranking: risk at matched coverage is identical for both (0.060, 0.097 and 0.152 at 50, 70 and 90%), so the reference answers more images under a coverage target and fewer under a risk target; Cal-B therefore remains the benchmark protocol. The certificate cut disease answered normal to about 4%, above its 2% target. These are empirical frequencies over related partitions, not distribution-free guarantees for a new site.

7 Discussion and Limitations

Why does low capacity work here? Fine-tuned ConvNeXt-T and Swin-T models (with or without DG losses) reached 0.85–0.89 on validation but G recall of 0.40–0.44 on test, yet their features hold no more site information than frozen ones (site probe 0.93–1.00 everywhere; Suppl. S6). The difference lies in how much the classifier exploits it: against fine-tuning, restraint in capacity is the likely cause; against a linear probe, token input and regularization are (Sec. 6.2). This is consistent with, not proof of, the account.

Label scope is a hidden shift. Because DR-only sources never assessed glaucoma, their "normal" images may include glaucoma. With the initial design, treating such labels as the set {N, G} raised development recall G (0.764 to 0.811), but at new sources

it over-called G (normal read as G: 0.248 to 0.443) and lowered bAcc (0.771 to 0.761; Suppl. S8). Label scope is a real error source, but discarding negative evidence is no remedy.

Residual artefact and deployment. With the initial design, normal images were read as glaucoma more often when the optic disc lay on the left, at all six development sites with normal images; mirrored training removed the gap but lowered recall G (0.768 to 0.652; Suppl. S8). Pooled scores hid a per-source range of 0.622–0.919 for LOTUS, so sites should measure per-class performance locally; 10 to 100 labelled target images restored conformal coverage near 90% on average (Suppl. S7).

Limitations. (1) All evidence is from public collections and the rotations reuse the same 14 sources; an independent, prospective cohort is needed. (2) Labels are used as published: glaucoma reference standards range from clinical work-up to image-based grading of suspects, D pools all DR grades, and test prevalence exceeds screening prevalence, so screening PPV would be lower; label error is unknown (Suppl. S1, Suppl. S8). (3) Glaucoma relies on two development sources, all development sources are Asian, HYGD and PALM hold one class, and HRF and AFIO are small. (4) Age and sex exist only for PAPILA, so fairness could not be assessed. (5) Pretraining overlap, including MedSigLIP's, was checked only against published descriptions; baselines were selected by in-distribution validation, with DG losses on ConvNeXt-T and DINOv2 only. (6) The configuration of LOTUS was chosen among ten development-qualified candidates on the published split and simplified after the rotation ablation; the bounded candidate range, the region hold-out and Chákṣu mitigate but do not remove this. (7) LOTUS is not a clinical tool.

8 Conclusion

Validation scores, calibrated probabilities and development-tuned thresholds do not reliably carry over to unseen fundus sources, and glaucoma, with thin site support, suffers most. LOTUS, leave-one-source-out linear heads on frozen tokens fused with zero-shot text, was more accurate and better calibrated than every trained baseline over 50 source rotations and the most accurate under region hold-out; text fusion and a regularized token head, not explicit site removal, carry that gain. Leave-one-source-out thresholds made risk promises hold far more often. Fundus benchmarks should report per source and class, check promises at unseen sources and test over source rotations; an independent cohort is the next step.

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